

Muhammad Shaheer Luqman

Driven Software & Applied AI Engineer building production-grade AI systems and end-to-end solutions for real-world industrial problems. Strong in problem-solving, system design, computer vision, NLP, and full-stack development.

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🐙 github.com/ShiheerLuqman | 🌐 muhammad-shaheer-portfolio.vercel.app

Experience

Applied Research Engineer (Full-Time) — Retrocausal Jun 2025 – Present

Full-Time • US-based (Remote)

Building **production-grade AI systems** and end-to-end solutions for industrial use cases, contributing to in-house frontend/backend software, and researching robot manipulation and Vision-Language-Action systems.

- ▶ Delivered a complete AI-powered solution for **BMW**, building frontend, backend, and AI components end-to-end.
- ▶ Worked on in-house software for both **frontend and backend**, supporting a range of development tasks.
- ▶ Supported deployments using various **AWS** products and **RunPod** infrastructure.
- ▶ Conducted **research** on unsupervised action segmentation, temporal long-video understanding, robot manipulation, and reinforcement learning, **publishing two papers** (see *Research & Publications*).

Python PyTorch Computer Vision VLMs Full-Stack Vue.js Django AWS RunPod Reinforcement Learning Robotics

Management Trainee Officer (MTO) — Confinality Jul 2025 – Oct 2025

Contract • Remote

Completed the **MTO** program while contributing to **NLP and speech-focused AI** workflows in production.

- ▶ Selected in a highly competitive MTO cohort.
- ▶ Built and supported **NLP** tasks including **sentiment analysis, speech detection, and transcription** pipelines.
- ▶ Worked on **voice generation** systems and **AI calling agent** creation.

NLP Voice Generation Sentiment Analysis Speech Detection Transcription AI Calling Agents

Applied Research Engineer (Intern) — Retrocausal Mar 2024 – Jun 2024

Internship • US-based (Remote)

Contributed to **industrial-scale AI systems** across computer vision, NLP, and LLM workflows.

- ▶ Built an end-to-end video activity recognition pipeline using **Body Pose Estimation + Vision-Language Models**, translating visual and textual data into PMTS codes for industrial process optimization.
- ▶ Developed **LLM-based prompt engineering** systems with automated Excel report generation and visual analytics.

Python Django Computer Vision LLMs Backend MediaPipe Excel Automation

Education

BS Computer Science — FAST NUCES Sep 2021 – Aug 2025

CGPA: 3.90 / 4.00

- ▶ Summa Cum Laude Honors.
- ▶ Rector's List (4.0 GPA) — 4 semesters.
- ▶ Dean's List (3.5+ GPA) — 4 semesters.

A Levels — Generation's School

Grades: A*, 2A

O Levels — Habib Public School

Grades: 4A*, 2A, 3B

Research & Publications

Unsupervised Skeleton-Based Action Segmentation via Hierarchical Spatiotemporal Vector Quantization 2026

arXiv 2026

A hierarchical spatiotemporal vector quantization framework for unsupervised skeleton-based temporal action segmentation, achieving SOTA on the HuGaDB, LARA, and BABEL benchmarks. [View on arXiv](#)

Enhancing Human-Likeness in Reinforcement Learning Agents via Hierarchical Macro Action Quantization 2026

arXiv 2026

A human-like RL framework that encodes human demonstrations into macro actions via a two-level hierarchical macro action quantization approach (HiMAQ), applied to dexterous control of a humanoid robot hand. On the D4RL benchmarks it outperforms the non-hierarchical MAQ baseline on human-likeness while matching or exceeding success rates, generalizing across IQL, SAC, and RLPD. [View on arXiv](#)

Ensuring Child Safety: An IoT-Based Surveillance System for Remote Monitoring and Detection of Anomalous Behavior

2024

ICETST 2024 • IEEE Xplore

Proposed an intelligent IoT-based system for remote child monitoring, combining hardware sensors, software components, and AI-driven anomaly detection to deliver real-time safety alerts. [View on IEEE Xplore](#)

Research Interests: Robotics, Reinforcement Learning, Computer Vision, Temporal Video Understanding, Vector Quantization, IoT + AI Systems, Vision-Language Models.

Projects

PixelForge — Digital Marketing Tool Using GenAI

Final Year Project ([GitHub](#))

A generative AI platform for marketing content (images/video), built for real-world workflows.

- Built a full-stack system using **React + Flask**, integrating **Stable Diffusion** for AI image generation.
- Developed pipelines for automated ad content generation (image/video).

Crypto Trading Simulator

Personal ([GitHub](#))

- Built a simulated crypto-trading platform on the **MERN** stack with real-time market-data APIs.
- Designed an intuitive UI using Ant Design and implemented database-backed simulation logic.

LushWear-IMS — Inventory Management System

Personal ([GitHub](#))

- Built backend APIs in **FastAPI** for products, search, and stock movement.
- Developed a desktop **Electron** frontend with a modern dark UI, integrating **Supabase** for persistence.

Fashion Industry Recommender System

Personal ([GitHub](#))

- A CNN-powered recommender using deep-learning visual embeddings to generate personalized item suggestions.
- Created a Python app interface to surface tailored product suggestions.

Contacts Organizer

Personal ([GitHub](#))

- A C++ contact manager with CRUD, search/sorting, and CSV-based persistence with backup handling.

Skills

AI / Machine Learning PyTorch, Scikit-learn, Hugging Face; Computer Vision (YOLO, MediaPipe); Diffusion Models (Stable Diffusion); NLP (BERT, Whisper, ILLabs); Reinforcement Learning (IQL, SAC, RLPD); Robotics & Robot Manipulation; LLM Prompt Engineering.

Software Engineering Backend (Django, Flask, FastAPI, Express.js); Frontend (React, Next.js, Vue); REST APIs, GraphQL; Postman; Databases (MySQL, MongoDB, Oracle SQL, PostgreSQL); Cloud (AWS, RunPod); System Design.

Systems & Tools Selenium (Web Automation), Arduino Programming, Assembly Language, IoT Systems, Git & Version Control, Linux, Docker.

Programming Languages Python, C / C++, JavaScript.

Honors & Awards

- **Gold Medal** — ICPC National Competitive Programming (Nov 2024)
- **Summa Cum Laude** — Graduated with a high GPA (3.90 / 4.00), FAST NUCES (2025)
- **Bronze Medal** — ICPC National Competitive Programming (Feb 2024)
- **Winner** — Code Sprint, PROCOM 2023 (2023)
- **Best Team Award** — Coder's Cup, ACM FAST
- **Rector's List (4.0 GPA)** — FAST NUCES, 4 semesters
- **Dean's List (3.5+ GPA)** — FAST NUCES, 4 semesters

Leadership & Volunteer Experience

- **Director Technical Affairs** — Developers Day'25
- **Technical Lead** — ACM NUCES
- **CS Competition Head** — Developers Day'24
- **CS Competition Co-Head** — PROCOM'24
- **Training & Mentorship Head** — ACM NUCES